



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS4130 – Computer Network Laboratory

Experiment No. 07

Switching and Virtual LANs (VLANs)

■ Objectives

- (a) Understand the purpose and benefits of VLANs in network design.
- (b) Create and configure VLANs on switches.
- (c) Implement inter-VLAN routing using:
 - Router-on-a-Stick (RoS) — using a router.
 - Switch Virtual Interfaces (SVIs) — using a Layer 3 switch.

■ Requirements

- (a) Cisco Packet Tracer software.
- (b) Two routers.
- (c) Three Layer 2 switches and one Layer 3 (multilayer) switch.
- (d) Thirteen PCs.

■ Pre-Lab Task

Students should review the fundamental concepts of switching and VLANs. In addition, they must build the network topology shown in Figure 3, including all notes, as outlined in Section 7.5.1. *Do not perform any configurations at this stage.*

7.1 Introduction

Switching is a fundamental concept in computer networks. It involves various operations performed by network *switches* and covers key topics such as Virtual Local Area Networks (*VLANs*), multilayer switching, trunking, switch architecture, and cabling structures.

In this experiment, we explore two approaches to implement VLANs:

- **Router-on-a-Stick (RoS):** Uses router sub-interfaces for inter-VLAN communication.
- **Switch Virtual Interface (SVI):** Utilizes a Layer 3 (multilayer) switch that provides virtual interfaces for inter-VLAN routing.

This experiment aims to help students understand VLANs and their role in enhancing network performance, security, and manageability. Through hands-on exercises in Cisco Packet Tracer, students will configure VLANs, assign ports, establish trunk links, and enable inter-VLAN communication using both router-based and switch-based methods.

7.1.1 How a Switch Works

A switch operates as a *plug-and-play device*, meaning it begins functioning immediately once powered on, even without manual configuration. Upon receiving a frame, it performs the following operations:

1. **Learns the source MAC address:** The switch extracts the source MAC address from the incoming frame and associates it with the corresponding port. This entry is added to the MAC address table if it does not already exist.
2. **Searches the MAC address table:** The switch looks up the destination MAC address in its table.
3. **Makes a forwarding decision:**
 - If the destination address *exists* in the table:
 - If it is on the *same port* from which the frame was received, the switch **drops** the frame.
 - Otherwise, it **forwards** the frame through the interface indicated in the table.
 - If the destination address *does not exist*, the switch **floods** the frame to all ports except the one it was received on.

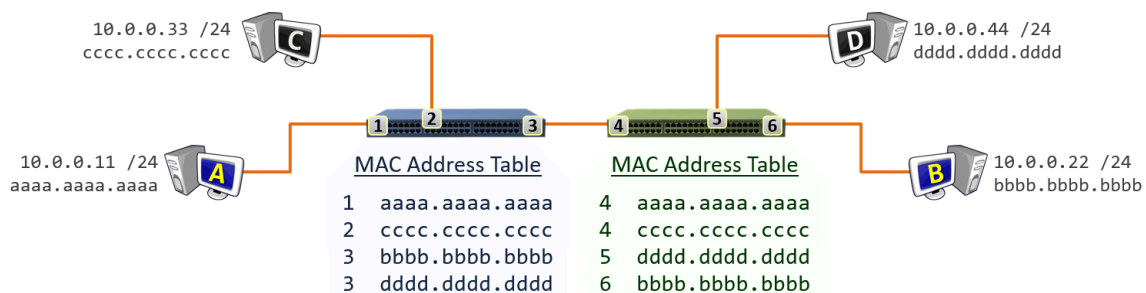


Figure 1: Switch MAC address table.

Over time, the switch builds a **MAC address table** mapping MAC addresses to specific ports, enabling efficient frame forwarding and reducing unnecessary traffic. Figure 1 illustrates the complete MAC address tables after all hosts have exchanged frames, allowing both switches to learn the location of each MAC address within the topology.

To display all dynamically learned MAC addresses, use the following command:

```
Switch# show mac address-table dynamic
```

7.1.2 Layer 3 Switches

A Layer 3 switch combines the capabilities of both Layer 2 (Data Link) and Layer 3 (Network) devices. It offers high-speed switching along with advanced routing functionalities, making it a powerful component in modern enterprise networks.

A) Key Features and Benefits:

- Typically equipped with 24 Ethernet ports, focusing on internal network connectivity rather than WAN interfaces.
- Operates at both Layer 2 and Layer 3, supporting switching and routing in one device.
- Supports IP routing, allowing forwarding decisions based on source and destination IP addresses.
- Performs inter-VLAN routing internally, eliminating the need for a separate router and improving performance.
- Simplifies configuration, management, and security.
- Reduces broadcast traffic and enhances fault isolation via effective segmentation.
- Provides high-speed scalability, low latency, and flow accounting for improved data transmission.
- Facilitates easier troubleshooting compared to traditional Layer 2 switches.

B) Disadvantages:

- More expensive than traditional Layer 2 switches.
- Better suited for enterprise environments; less ideal for small networks.
- Lacks WAN functionality, requiring additional devices for external connectivity.
- Offers less routing flexibility than dedicated routers in complex network architectures.

7.2 Virtual Local Area Networks (VLANs)

A **VLAN** is a logical grouping of devices within a network that allows them to communicate as if they were on the same physical LAN, even when they are located on different segments. VLANs are *configured through software*, providing flexibility, manageability,

and enhanced security. They allow logical separation of network segments without additional physical infrastructure—so when a device is moved, it can remain in the same VLAN without hardware changes.

7.2.1 When to Use VLANs

VLANs are especially useful for logically *segregating traffic at Layer 2* (the data link layer). Common scenarios include:

- Assigning each IP subnet to its own VLAN.
- Separating departments or services within an organization.
- Reducing broadcast traffic by limiting the broadcast domain.
- Preventing unauthorized access between subnets (a device cannot join another subnet by simply changing its IP address).

7.2.2 IEEE 802.1Q VLAN Standard

IEEE 802.1Q is the industry standard protocol for **VLAN tagging** in Ethernet networks. It enables multiple VLANs to coexist on a single physical link by inserting a **4-byte VLAN tag** into the Ethernet frame header—between the *source MAC address* (48-bit field) and the *EtherType* field (which normally identifies the upper-layer protocol, for example, IPv4 = 0x0800, ARP = 0x0806, IPv6 = 0x86DD), as illustrated in Figure 2.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18...
Destination address						Source address						VLAN tag				EtherType	Payload	
												TPID		TCI				

Figure 2: 802.1Q VLAN frame structure.

The VLAN tag consists of the following components:

- **Tag Protocol Identifier (TPID):** Always 0x8100, indicating that the frame carries VLAN information.
- **Tag Control Information (TCI):** A 16-bit field containing:
 - **Priority Code Point (PCP):** 3 bits for Quality of Service (QoS) priority.
 - **Drop Eligible Indicator (DEI):** 1 bit indicating whether the frame can be dropped during congestion.
 - **VLAN Identifier (VID):** 12 bits identifying the VLAN (range: 0–4095).
 - * **VID 0:** Reserved to indicate that the frame carries only priority information (no VLAN membership).
 - * **VID 1:** Default VLAN assigned to unconfigured ports (should be avoided in production for better segmentation and security).
 - * **VIDs 2 – 4094:** Available for general use (actual supported number depends on hardware).

- * VID **4095**: Reserved and not used for normal traffic (typically reserved for implementation-specific or special control purposes).

7.2.3 Creating a VLAN

To create a specific VLAN on a switch, use the following command:

```
Switch(config)# VLAN <VLAN-NUMBER>
```

You can verify that the VLAN has been configured by running:

```
Switch# show VLAN
```

7.3 Trunk and Access Ports

When working with VLANs, *trunk* and *access* ports are essential for managing VLAN traffic effectively.

A) Trunk Ports: A trunk port carries traffic for *multiple* VLANs between devices using **IEEE 802.1Q tagging**. Trunk ports are commonly used in the following scenarios:

- Between two switches
- Between a router and a switch (RoS setup)
- Between a Layer 3 switch and a Layer 2 switch (SVI setup)

To configure a **Layer 2 switch** port as a trunk, access the desired interface and execute the following command:

```
Switch(config-if)# switchport mode trunk
```

On the other hand, when configuring a **Layer 3 switch** port to operate in trunk mode, 802.1Q encapsulation must be explicitly specified before enabling trunking as follows:

```
Switch(config-if)# switchport trunk encapsulation dot1q  
Switch(config-if)# switchport mode trunk
```

Notes:

- When one end of a link is configured as a trunk, the other end typically switches to trunk mode automatically.
- When connecting a Layer 2 switch to a Layer 3 switch, only the Layer 3 side needs encapsulation configuration. The Layer 2 switch only needs to be set to trunk mode, as it uses 802.1Q tagging by default and does not require manual encapsulation specification.

B) Access Ports: Access ports carry traffic for a **single** VLAN and handle **untagged** frames from connected devices. To assign a port to a specific VLAN, access the interface and execute:

```
Switch(config-if)# switchport access VLAN <VLAN-NUMBER>
```

To configure multiple *consecutive interfaces* at once, you can define a range of interfaces and apply the VLAN assignment to all of them:

```
Switch(config)# interface range <TYPE>
                  <SLOT>/<START-PORT> - <END-PORT>
Switch(config-if-range)# switchport access VLAN
                        <VLAN-NUMBER>
```

Note:

A switch port can be configured to act as an access port for one VLAN and a trunk for others, sending both tagged and untagged frames. However, this mode increases the risk of *VLAN hopping attacks* and is not recommended.

7.4 Inter-VLAN Routing

Each VLAN is a separate network; therefore, devices in different VLANs require routing to communicate. This can be achieved using:

- Router-on-a-Stick (RoS): A router interface is divided into multiple sub-interfaces, each handling traffic for one VLAN.
- Layer 3 Switch – Switch Virtual Interface (SVI): Routing handled directly by the switch using virtual interfaces.

7.4.1 Router-on-a-Stick (RoS) and Sub-Interfaces

A **sub-interface** is a logical interface created under a physical router interface. Each sub-interface operates independently, with its own IP address and VLAN association. The main physical interface *does not* need an IP address when sub-interfaces are used.

A) Creating a Sub-Interface

To define a sub-interface on a router's physical interface, use the following command:

```
Router(config)# interface <TYPE> <SLOT>/<PORT>.
                  <SUB-INTERFACE-NUMBER>
Router(config-subif)#
```

- <TYPE>: Interface type (e.g., FastEthernet, GigabitEthernet)
- <SLOT>/<PORT>: Physical interface designation
- <SUB-INTERFACE-NUMBER>: Identifier for the logical sub-interface (often matching the VLAN ID)

B) Assigning VLAN and IP Address to a Sub-Interface

```
Router(config-subif)# encapsulation dot1Q <VLAN-ID>
Router(config-subif)# ip address <IP-ADDRESS> <SUBNET-MASK>
```

- **encapsulation dot1Q <VLAN-ID>**: Associates the sub-interface with a specific VLAN using IEEE 802.1Q tagging.
- **ip address <IP-ADDRESS> <SUBNET-MASK>**: Assigns an IP address to the sub-interface, enabling Layer 3 communication for the corresponding VLAN.

7.4.2 Layer 3 Switch and Switch Virtual Interfaces (SVIs)

To enable inter-VLAN routing on a Layer 3 switch, several configuration steps are required. These include converting switch ports to routed ports, enabling IP routing, creating VLANs, and assigning IP addresses to the corresponding SVIs.

A) Converting a Switch Port to a Routed Port

To configure a Layer 3 switch port to function like a router interface (i.e., to route traffic), you must disable Layer 2 switching on that port and assigning it an IP address:

```
Switch(config-if)# no switchport
Switch(config-if)# ip address <IP-ADDRESS> <SUBNET-MASK>
```

B) Enabling IP Routing

By default, Layer 3 switches do not perform routing. To enable this functionality, use the following command in global configuration mode:

```
Switch(config)# ip routing
```

C) Configuring Switch Virtual Interfaces (SVIs)

Each VLAN requires an SVI that serves as the *default gateway* for hosts within that VLAN. The switch uses these virtual interfaces to route traffic between VLANs.

To configure an SVI, use the following commands:

```
Switch(config)# interface vlan <VLAN-NUMBER>
Switch(config-if)# ip address <IP-ADDRESS> <SUBNET-MASK>
```

7.5 Procedure

In this lab, we will build a network consisting of two routers, one multilayer switch, three Layer 2 switches, and several PCs distributed across multiple networks and VLANs.

1. **Router 0** will host VLANs **10**, **20**, and **30**.
2. **Router 1** will host VLANs **40** and **50**.
3. **Multilayer Switch 0** will host VLANs **60** and **70**.

To enable inter-VLAN communication between devices connected to different routers and the multilayer switch, we will configure the *Open Shortest Path First (OSPF)* routing protocol in **Area 0** on all routers and the multilayer switch. This lab demonstrates VLAN segmentation, sub-interface configuration, SVI configuration, and dynamic routing between VLANs.

7.5.1 Building the Topology

Construct the network topology shown in Figure 3, which includes the following devices:

- Routers: Use **Router-PT**.
- Layer 2 Switches: Use **Switch-PT**.
- Multilayer Switch: Use **3560-24PS**.
- PCs: Use **PC-PT**.
- Connections: Use the “**Automatically Choose Connection Type**” option to connect PCs, switches, and routers.

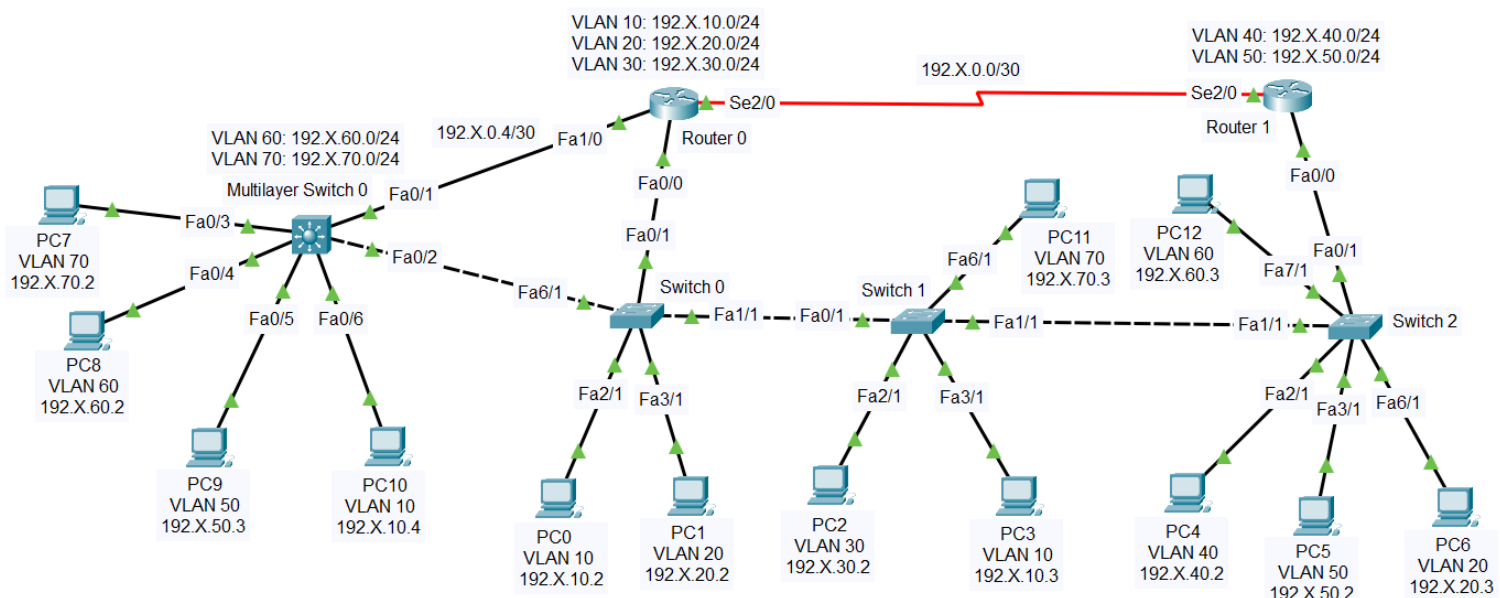


Figure 3: Switching and VLANs topology.

For **Switch-PT**, you must physically add an additional interface(s) using the **PT-SWITCH-NM-1CFE Module**, as illustrated in Figure 4.

1. Click on the switch and open the *Physical* tab.

2. Turn off the switch using the power switch.
3. Locate the **PT-SWITCH-NM-1CFE** module.
4. Drag and drop it into the empty slot on the switch.
5. Turn the switch back on using the power switch.

Note:

Use the **write** command (e.g., **Switch# write**) to permanently save your configurations before powering off any network device (router, switch, etc.).

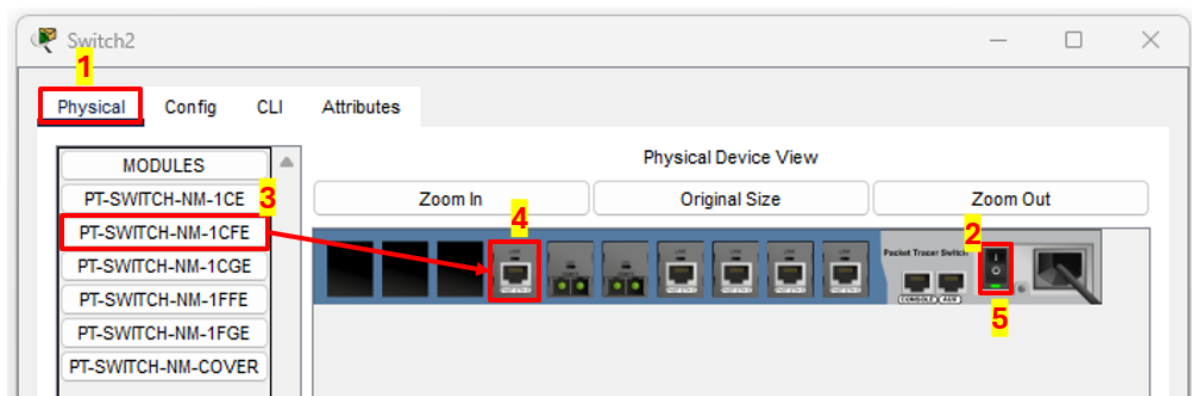


Figure 4: Adding an extra interface to a switch.

Assign appropriate IP addresses to all PCs and routers, ensuring that each device receives a unique address within the valid host range of its respective network. The value of **X** in the network address corresponds to the *last two digits* of your student ID. For example, if the network address is **192.X.40.0/24** and your student ID is **1230105**, then **X = 05**, resulting in the network address **192.5.40.0/24**.

7.5.2 Configuring Routers

A) Configuring Routers Sub-Interfaces

When implementing *Router-on-a-Stick* for VLAN configuration, a sub-interface must be created for each VLAN hosted on the router. Each sub-interface acts as the **default gateway** for its corresponding VLAN. For example, to configure sub-interface **Fa0/0.10** on **Router 0** for **VLAN 10**, follow these steps:

1. Enable the physical interface (no IP address is required since sub-interfaces will be used):

```
Router(config)# interface Fa0/0
Router(config-if)# no shutdown
```

2. Create and configure the sub-interface for VLAN 10:

```
Router(config)# interface Fa0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.X.10.1 255.255.255.0
```

Now, complete the configuration by assigning IP addresses for:

- **VLAN 20 and VLAN 30 on Router 0.**
- **VLAN 40 and VLAN 50 on Router 1.**

B) Configuring OSPF Routing

Next, enable and configure **OSPF (Area 0)** on both **Router 0** and **Router 1**, following the procedure described in *EXP. No. 4, Section 4.3.3*. For example, to configure OSPF on **Router 0**, use the following commands:

```
Router(config)# router ospf 1
Router(config-router)# network 192.X.0.0 0.0.0.3 area 0
Router(config-router)# network 192.X.0.4 0.0.0.3 area 0
Router(config-router)# network 192.X.10.0 0.0.0.255 area 0
Router(config-router)# network 192.X.20.0 0.0.0.255 area 0
Router(config-router)# network 192.X.30.0 0.0.0.255 area 0
```

Apply a similar configuration on **Router 1**, modifying the **network** statements to match its directly connected networks.

7.5.3 Configuring Layer 2 Switches

A) Creating VLANs

To create **VLAN 10** on **Switch 0**, use the following commands:

```
Switch(config)# VLAN 10
Switch(config-vlan)# exit
```

Note:

Since a packet tagged with a specific VLAN can traverse multiple switches in the network, a switch will discard any packet belonging to a VLAN that is not configured on it. Therefore, you must ensure that all switches are configured with the complete set of VLANs.

Create **VLANs 20, 30, 40, 50, 60, and 70** on **Switch 0**, and repeat the same process to configure the VLANs on **Switch 1** and **Switch 2**.

B) Configuring Switch Access Ports

To assign an interface to an existing VLAN, access the desired port and apply the **access VLAN** command. For example, to assign port **Fa2/1** on **Switch 0** to **VLAN 10**, use the following commands:

```
Switch(config)# interface Fa2/1
Switch(config-if)# switchport access VLAN 10
```

Next, configure the interfaces connecting each PC to the switches as access ports for their corresponding VLANs on **Switch 0**, **Switch 1**, and **Switch 2**, following the topology diagram and the summary in Table 1.

Note:

If you attempt to assign a VLAN that has not yet been created, the switch will automatically create it.

Table 1: Switch port configuration summary.

Switch	Port(s)	Mode	VLAN
Switch 0	Fa0/1, Fa1/1, Fa6/1	Trunk	—
	Fa2/1	Access	VLAN 10
	Fa3/1	Access	VLAN 20
Switch 1	Fa0/1, Fa1/1	Trunk	—
	Fa2/1	Access	VLAN 30
	Fa3/1	Access	VLAN 10
	Fa6/1	Access	VLAN 70
Switch 2	Fa0/1, Fa1/1	Trunk	—
	Fa2/1	Access	VLAN 40
	Fa3/1	Access	VLAN 50
	Fa6/1	Access	VLAN 20
	Fa7/1	Access	VLAN 60
Multilayer Switch 0	Fa0/2	Trunk	—
	Fa0/3	Access	VLAN 70
	Fa0/4	Access	VLAN 60
	Fa0/5	Access	VLAN 50
	Fa0/6	Access	VLAN 10

C) Configuring Switch Trunk Ports

To configure an interface as a trunk port, access the port and apply the trunk mode command. For example, to configure port **Fa1/1** on **Switch 0** as a trunk, use:

```
Switch(config)# interface Fa1/1
Switch(config-if)# switchport mode trunk
```

Note:

When one end of a link is configured as a trunk, the other end typically transitions to trunk mode automatically. In this example, configuring **Fa1/1** on **Switch 0** will also cause **Fa0/1** on **Switch 1** to operate as a trunk.

Configure all links between switches, as well as links between routers and switches, to operate in trunk mode on **Switch 0**, **Switch 1**, and **Switch 2**.

7.5.4 Configuring the Multilayer Switch**A) Converting a Switch Port to a Routed Port**

To assign an IP address to the interface connecting the multilayer switch to the router, the switch port must first be converted to a routed port. After that, you can assign it

an IP address. For example, to configure IP address 192.X.0.5/30 on Fa0/1, use the following commands:

```
Switch(config)# interface Fa0/1
Switch(config-if)# no switchport
Switch(config-if)# ip address 192.X.0.5 255.255.255.252
```

B) Creating VLANs

To create **VLAN 70** on the multilayer switch, enter:

```
Switch(config)# vlan 70
Switch(config-vlan)# exit
```

Because packets tagged with a VLAN may pass through the multilayer switch, the switch will drop such packets if the corresponding VLAN has not been created. Therefore, **you must also create VLANs 60, 50, 40, 30, 20, and 10 on the multilayer switch** to ensure proper VLAN handling.

Note:

Each switch—including the multilayer switch—must have all VLANs created to ensure correct and seamless operation across the network.

C) Configuring VLAN Interfaces (Switch Virtual Interfaces – SVIs)

To enable the multilayer switch to act as the default gateway for **VLANs 60 and 70**, you must configure their respective SVIs. For example, to configure **VLAN 60** with IP address 192.X.60.1/24, use:

```
Switch(config)# interface vlan 60
Switch(config-if)# ip address 192.X.60.1 255.255.255.0
```

Next, configure the SVI for VLAN 70 in a similar manner.

D) Enabling Routing and Configuring OSPF

By default, routing is disabled on a Layer 3 switch. To enable it, use:

```
Switch(config)# ip routing
```

After enabling routing, configure the **OSPF routing protocol (Area 0)** on the multilayer switch using the following commands:

```
Switch(config)# router ospf 1
Switch(config-router)# network 192.X.60.0 0.0.0.255 area 0
Switch(config-router)# network 192.X.70.0 0.0.0.255 area 0
Switch(config-router)# network 192.X.0.4 0.0.0.3 area 0
```

These commands start **OSPF process 1** and advertise the networks corresponding to **VLAN 60, VLAN 70**, and the point-to-point link connecting the multilayer switch to **Router 0**.

E) Configuring the Trunk Port

To allow VLAN traffic between **Multilayer Switch 0** and **Switch 0**, configure Fa0/2 on the multilayer switch as a trunk port:

```
Switch(config)# interface Fa0/2
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport mode trunk
```

Note:

When one end of a link is configured as a trunk, the other end typically transitions to trunk mode automatically. Therefore, if **Switch 0**'s Fa6/1 port is already configured as a trunk, you only need to manually configure port Fa0/2 on **Multilayer Switch 0** to specify the encapsulation.

F) Configuring Access Ports

To assign an interface to a VLAN, access the port and apply the access VLAN command. For example, to assign Fa0/3 on the multilayer switch to **VLAN 70**, use:

```
Switch(config)# interface Fa0/3
Switch(config-if)# switchport access vlan 70
```

Finally, configure the access ports for all PCs connected to the multilayer switch according to their respective VLAN assignments.

7.6 VLAN Communication Path Analysis

Based on the network topology illustrated in Figure 3, the following outlines the communication paths for various VLAN and routing scenarios. You may use **Packet Tracer's Simulation Mode** to verify these scenarios.

A) Communication Within the Same VLAN Across Switches (PC0 to PC3)

Both PC0 and PC3 are in VLAN 10 (192.X.10.0/24) and do not require a router for communication, as they share the same VLAN.

- PC0 sends an Ethernet frame to PC3's MAC address.
- The frame travels from PC0 to Switch 0 via access port Fa2/1.
- From Switch 0, the frame moves through the trunk port Fa1/1 to Switch 1.
- Switch 1 receives the frame and forwards it directly to PC3 via its access port Fa3/1 (also assigned to VLAN 10).
- No routing occurs since this is Layer 2 communication within the same VLAN.

Path: *PC0 → SW0 → SW1 → PC3*

B) Communication Between Different VLANs with the Default Gateways on the Same Router (PC0 to PC6)

PC0 (VLAN 10, IP: 192.X.10.2) is trying to reach PC6 (VLAN 20, IP: 192.X.20.3). Since both VLAN 10 and VLAN 20 are routed by Router 0, inter-VLAN routing occurs via its sub-interfaces on Fa0/0.

- PC0 sends the packet to its default gateway (Router 0 via sub-interface Fa0/0.10).
- The packet travels from PC0 to Switch 0 (access port Fa2/1), then through the trunk port Fa0/1 to Router 0.
- Router 0 receives the packet on sub-interface Fa0/0.10, performs inter-VLAN routing, and forwards it out via sub-interface Fa0/0.20 (for VLAN 20).
- The routed packet goes back to Switch 0 via trunk port Fa0/1.
- From Switch 0, it travels through trunk links to Switch 1 and then to Switch 2.
- Finally, it is delivered to PC6 via its access port on Switch 2.

Path: *PC0* → *SW0* → *R0* → *SW0* → *SW1* → *SW2* → *PC6*

C) Communication Between Different VLANs with the Default Gateways on Different Routers (PC0 to PC5)

PC0 (VLAN 10, IP: 192.X.10.2) and PC5 (VLAN 50, IP: 192.X.50.2) are in different VLANs that are routed through different routers — Router 0 handles VLAN 10 and Router 1 handles VLAN 50.

- PC0 sends the packet to its default gateway (Router 0 via sub-interface Fa0/0.10).
- The packet travels from PC0 to Switch 0 (access port Fa2/1), and then to Router 0 via the trunk connection.
- Router 0 receives the packet on Fa0/0.10, identifies that the destination is in a remote VLAN (VLAN 50), and routes the packet out via the serial interface Se2/0 to Router 1.
- The packet reaches Router 1 through the point-to-point serial link (192.X.0.1 → 192.X.0.2).
- Router 1 receives the packet and forwards it via sub-interface Fa0/0.50, associated with VLAN 50.
- The packet travels from Router 1 to Switch 2 through the trunk port.
- Finally, Switch 2 forwards the packet to PC5 through the access port on VLAN 50.

Path: *PC0* → *SW0* → *R0* → *R1* → *SW2* → *PC5*

Additional Analysis Tasks

Carefully describe the communication paths for each of the following scenarios. You may use **Packet Tracer's Simulation Mode** to verify your answers. Then, discuss your findings with your instructor or TA.

1. Communication path **from PC8 to PC12**.
2. Communication path **from PC7 to PC8**.
3. Communication path **from PC9 to PC10**.
4. Communication path **from PC3 to PC11**.

7.7 ToDo

This section will be provided by the instructor.